1 Qiskit program
parse
2 srcML, line 7
interpret
3 Operation trace
slice
4 Slice of q_0
4 circuit = QuantumCircuit(2)
5 circuit.h(0)
6 circuit.h(1)
fault
7 circuit.cx(0,1)
8 circuit.measure_all()
lines 10–16: run on a simulator, print
Expected (|01⟩+|10⟩)/√2; line 6 should apply X.
expr_stmt pos="7:1"
call
cx: control q_0, target q_1
wire rule: writes both
name
circuit.cx
argument_list
argument
0
argument
1
line 5
line 6
line 7
criterion
{q_0}
q_0
H
kickback:
cx writes q_0
q_1
H
shaded: kept by the wire rule's backward pass from q_0. The terminal measurement of line 8 is outside the criterion.
wire
PDG
4 circuit = QuantumCircuit(2)
5 circuit.h(0)
6 circuit.h(1)
7 circuit.cx(0,1)
8 circuit.measure_all()
wire rule: fault line 6 kept
classical PDG: line 5 only, fault missed
full square: quantum line; light: classical line (circuit construction).